FELIZBAY QUALITY ENGINEERING LABORATORY · STAGE 3 VERIFICATION REPORT (ORT / STRUCTURAL DESIGN)

Stage 3 Test Report: Full-Load Hydrostatic & Dynamic Impact Validation Report

Deep-Water Lateral Rigidity, Wave Impact Fatigue & Fitting Flange Validation for Felizbay Pool (400 cm × 300 cm × 120 cm, 20 cm Wall)

1. Structural Parameters, Test Bay Setup & Load Conditions

Report No.	`RPT-FB-QC-20261113-S3`	**Governing SOP**	`WI-QC-DS-2026-001 (V1.0)` Stage 3
Brand / Model	**Felizbay** Large Rectangular Drop-Stitch Pool (`FB-DSP-400`)	**Outer / Inner Size**	Outer **400 × 300 × 120 cm** / Inner Basin 360 × 260 cm
Wall & Bonding	**20 cm** Thick Drop-Stitch Wall / **Heat Weld + PU Adhesive**	**Air & Water Load**	**P_w = 8.0 PSI** / Rated 105 cm (9,828 L), 110% Overload 10,810 L
Sample Batch / Unit	Batch `LOT-202611-A08` / **Unit #3** (Structural & Fitting Destruction)	**Corner & Base Design**	R160 mm Radiused Double-Patch Corners + Perimeter Triangular Gusset
Test Facility	Heavy Structural Testing Bay (Level Concrete + 5 mm Mat + Gantry)	**Duration / Result**	72 h Continuous Hydrostatic + Dynamic Impact / **Stage 3 Approved (PASS)**

Test Scenario & Structural Focus: With outer dimensions of **400 cm × 300 cm × 120 cm**, **Felizbay Unit #3** holds ~10 metric tons of water at full capacity, generating > 11.3 kPa hydrostatic floor pressure. Executed in the Structural Testing Bay with an overhead gantry rig, Stage 3 validates the bending stiffness of the **20 cm thick drop-stitch wall**, fatigue endurance of the heat-welded and PU-bonded corners, and mechanical retention of valve/drain flanges.

2. Stage 3 Structural Testing Bay & Component Destruction Photographs





Fig. 3-1: Full-Load Hydrostatic, Sandbag & Pendulum Rig (Tests 3.1 & 3.2)
Unit #3 (400 × 300 × 120 cm) pressurized to 8.0 PSI and filled with water; weighted sandbags test top-rail buckling while a gantry-suspended cylindrical bag swings from a 1.0 m offset.

Fig. 3-2: 20 cm Thick-Wall Valve Flange Torque Destruction Test (Test 3.3)
At 8.0 PSI internal pressure, a QC engineer engages the custom slotted spanner and digital torque wrench on the valve flange, applying clockwise and counter-clockwise torque to failure.

3. Detailed Test Parameters & Measured Data Records

3.1 Test 3.1: 110% Overloaded Water Filling & 72-Hour Hydrostatic Lateral Deflection Test

Test Parameters: **Unit #3** placed on a level concrete floor (≤ 3 mm/2m flatness) with a 5 mm anti-slip mat, inflated to $\mathbf{P_w = 8.0\ PSI}$, and filled to **110%** of rated water height (115.5 cm depth, 4.5 cm below top rim, **10,810 L** total). Held for **72 hours** (pressure trimmed twice daily to 8.0 PSI); laser level and digital inclinometer tracked top horizontal displacement ΔX and outward tilt angle θ (Spec: θ ≤ 10°, zero stress-whitening or peeling on bottom inner seam after draining).

Elapsed Time	Long-Wall Top Shift ΔX (mm)	Short-Wall Top Shift ΔX (mm)	Max Outward Tilt θ (Spec ≤ 10°)	20 cm Wall Profile & Spill Status	Result
0h (110% Load)	31.5 mm	18.4 mm	3.0°	20 cm wall upright without waisting; 45 mm freeboard, 0 spill	Normal
24h	46.8 mm	27.2 mm	4.5°	Lateral creep under 10.81 t water load stabilizing, no folding	Normal
48h	54.2 mm	31.6 mm	5.2°	Uniform gentle outward bowing at mid-span; zero water overflow	Normal
72h (Final)	**58.4 mm**	**34.2 mm**	**5.6°**	**Tilt 5.6° ≤ 10°; post-drain inspection confirmed bottom weld+glue L-seam & triangular gusset intact with zero whitening**	**PASS**

3.2 Test 3.2 (150 kg Top Buckling & 500-Cycle Pendulum Impact) & Test 3.3 (Fitting Destruction)

Test 3.2 Parameters: At 100% rated water (9,828 L) and 8.0 PSI, **3 × 50 kg = 150 kg** sandbags placed on a 600 × 150 mm rubber pad at mid-long-wall top for **60 min**; next, a **Φ300 mm / 50 kg** cylindrical bag aligned at waterline (2/3 wall height) swung from a **1.0 m** horizontal offset for 250 mid-wall + 250 corner impacts (**500 cycles** total, 18 cycles/min), followed by $\mathbf{1.5 \times P_w = 12.0\ PSI}$ hold for 60 min with soap-solution leak check.
Test 3.3 Parameters: At 8.0 PSI, custom spanner and digital torque wrench twisted inflation and drain valve bases at **5°/s** CW/CCW to failure; post-drain 400 × 400 mm drain-port cutout pulled axially at **100 mm/min** to failure.

Test Sub-Item	Factory Acceptance Criteria	Measured Peak Data & Failure Mode Observation	Result
3.2a Top 150 kg Buckling	60 min load without wall folding/buckling; full elastic recovery	Max top depression **11.8 mm** over 60 min with zero buckling; recovered 100% within 2 min of unloading	**PASS**
3.2b 500-Cycle Wave Impact	Post-impact 1.5×P_w (12 PSI) 1 h hold with 0 drop & 0 bubbles	12.0 PSI 60 min hold showed **0.0 PSI** drop; 0 bubbles across R160 mm heat-welded + PU-bonded corner patches	**PASS**
3.3a Valve Base Torque	Ultimate failure torque **≥ 25 N·m**, zero flange peel	Inflation valve CW/CCW: **36.4 / 35.1 N·m**; Drain valve: **32.8 / 31.9 N·m** (thread strip; weld+glue flange intact)	**PASS**
3.3b Drain Axial Pull-Out	Ultimate pull-out force **≥ 800 N**, zero flange delamination	Peak pull force **1,185 N** (+48.1% over spec); cap pull-ring snapped while flange skirt remained 100% bonded	**PASS**

4. Stage 3 Conclusion & Structural Design Evaluation

Stage 3 Summary Conclusion: The structural architecture of the **Felizbay** Drop-Stitch Inflatable Pool (`FB-DSP-400`, 400 × 300 × 120 cm)—combining a **20 cm thick drop-stitch wall**, full-perimeter internal triangular base gusset, and R160 mm double-layer heat-welded + PU-bonded corner patches—is fully validated. Under 110% water overload (10,810 L) for 72 h, outward wall tilt reached only **5.6°** (vs. 10° limit), 150 kg top loading produced zero buckling, 500 pendulum impacts yielded zero leaks, and fitting retention exceeded targets by > 30%.

Structural Technician: H. Chen (QC Lab)	**R&D / QA Reviewer**: M. Lin (Quality Eng.)	**Report Date**: 2026-11-13